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MAGNETIC TUNNEL JUNCTION WITH DUAL REFERENCE LAYERS HAVING PARALLEL MAGNETIZATION DIRECTIONS AND METHODS FOR OPERATING THE SAME

Abstract

A magnetoresistive memory cell includes a first electrode; a second electrode; and a layer stack located between the first electrode and the second electrode and comprising, from one end to another, a first reference layer, a first tunnel barrier layer, a free layer, a second tunnel barrier layer, and a second reference layer. A first one of the first reference layer and the second reference layer comprises a positive spin polarization material. A second one of the first reference layer and the second reference layer comprises a negative spin polarization material. A magnetization direction of the second reference layer is parallel to a magnetization direction of the first reference layer.

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Background/Summary

FIELD

[0001] The present disclosure relates generally to the field of magnetic memory devices and specifically to a magnetic memory device including a magnetic tunnel junction with dual reference layers having parallel magnetization directions and methods for operating the same.

BACKGROUND

[0002] Spin-transfer torque (STT) refers to an effect in which the orientation of a magnetic layer in a magnetic tunnel junction or spin valve is modified by a spin-polarized current. Generally, electric current is unpolarized with electrons having random spin orientations. A spin polarized current is one in which electrons have a net non-zero spin due to a preferential spin orientation distribution. A spin-polarized current can be generated by passing electrical current through a magnetic polarizer layer. When the spin-polarized current flows through a free layer of a magnetic tunnel junction or a spin valve, the electrons in the spin-polarized current can transfer at least some of their angular momentum to the free layer, thereby producing a torque on the magnetization of the free layer.

[0003] When a sufficient amount of spin-polarized current passes through the free layer, spin-transfer torque can flip the magnetization orientation of the free layer. A resistance differential of a magnetic tunnel junction between different magnetization states of the free layer can be employed to store data within the magnetoresistive random access memory (MRAM) cell depending on whether the magnetization of the free layer is parallel or antiparallel to the magnetization of the reference layer.

SUMMARY

[0004] According to an embodiment of the present disclosure, a device comprising a magnetoresistive memory cell is provided. The magnetoresistive memory cell comprises: a first electrode; a second electrode; and a layer stack located between the first electrode and the second electrode and comprising, from one end to another, a first reference layer, a first tunnel barrier layer, a free layer, a second tunnel barrier layer, and a second reference layer, wherein: a first one of the first reference layer and the second reference layer comprises a positive spin polarization material; a second one of the first reference layer and the second reference layer comprises a negative spin polarization material; and a magnetization direction of the second reference layer is parallel to a magnetization direction of the first reference layer.

[0005] According to an embodiment of the present disclosure, a method of operating a spin transfer torque (STT) magnetoresistive random memory (MRAM) device includes providing a magnetoresistive memory cell which comprises a first electrode, a second electrode, and a layer stack located between the first electrode and the second electrode and comprising, from one end to another, a first reference layer, a first tunnel barrier layer, a free layer, a second tunnel barrier layer, a second reference layer, wherein the first reference layer and the second reference layer have magnetization directions that are parallel to each other; applying a read bias across the first electrode and the second electrode while the magnetization directions of the first reference layer and the second reference layer are parallel to each other; and determining a magnetization direction of the free layer relative to the magnetization directions of the first reference layer and the second reference layer based on a magnitude and a direction of electrical current that flows through the layer stack under the read bias voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a schematic diagram of a memory device including an array of magnetoresistive

memory cells according to an embodiment of the present disclosure.

[0007] FIG. 2A illustrates a magnetoresistive memory cell in a parallel free layer state according to an embodiment of the present disclosure.

[0008] FIG. 2B illustrates the magnetoresistive memory cell in an antiparallel free layer state according to an embodiment of the present disclosure.

[0009] FIG. 3A-3D are schematic illustrations of spin polarization changes for electrons passing through a spin polarization material in case the first reference layer and the second reference layer have upward magnetization directions.

[0010] FIG. 4A illustrates an alternative configuration of a magnetoresistive memory cell in a parallel free layer state according to an embodiment of the present disclosure.

[0011] FIG. 4B illustrates the alternative configuration of the magnetoresistive memory cell in an antiparallel free layer state according to an embodiment of the present disclosure.

[0012] FIG. 5A-5D are schematic illustrations of spin polarization changes for electrons passing through a spin polarization material in case the first reference layer and the second reference layer have downward magnetization directions.

DETAILED DESCRIPTION

[0013] Various spin-transfer torque magnetic random access memory (STT-MRAM) devices are known in the art. Some of the STT MRAM devices comprise dual (e.g., double) magnetic tunnel junction (MTJ) STT MRAM devices which contain two reference layers on opposite sides of the free layer. Such dual MTJ STT-MRAM devices include a first (e.g., top) reference layer having a magnetization direction that is antiparallel to the magnetization direction of a second (e.g., bottom) reference layer in order to provide additive STT to the free layer and thus reduce the current needed to switch the free layer magnetization direction. However, such dual MTJ STT-MRAM devices suffer from a write error rate limitation due to back-hopping of the first reference layer. This back-hopping problem is more severe when the free layer is in the parallel state in which the magnetization directions of the free layer and the second reference layer are parallel to each other, but the magnetization direction of the free layer is antiparallel to the magnetization direction of a second reference layer. The embodiments of the present disclosure are directed to a dual MTJ STT-MRAM device including a magnetic tunnel junction with dual reference layers having parallel magnetization directions (i.e., in parallel alignment) and methods for operating the same, the various aspects of which are described below.

[0014] The drawings are not drawn to scale. Multiple instances of an element may be duplicated where a single instance of the element is illustrated, unless absence of duplication of elements is expressly described or clearly indicated otherwise. Same reference numerals refer to the same element or to a similar element. Elements having the same reference numerals are presumed to have the same material composition unless expressly stated otherwise. Ordinals such as “first,” “second,” and “third” are employed merely to identify similar elements, and different ordinals may be employed across the specification and the claims of the instant disclosure. As used herein, a first element located “on” a second element can be located on the exterior side of a surface of the second element or on the interior side of the second element. As used herein, a first element is located “directly on” a second element if there exist a physical contact between a surface of the first element and a surface of the second element. As used herein, an “in-process” structure or a “transient” structure refers to a structure that is subsequently modified.

[0015] As used herein, a “layer” refers to a material portion including a region having a thickness. A layer may extend over the entirety of an underlying or overlying structure, or may have an extent less than the extent of an underlying or overlying structure. Further, a layer may be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer may be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer may extend horizontally, vertically, and/or along a tapered surface. A substrate may be a layer, may include one

or more layers therein, and/or may have one or more layer thereupon, thereabove, and/or therebelow.

[0016] As used herein, a “layer stack” refers to a stack of layers. As used herein, a “line” refers to a layer that has a predominant direction of extension, i.e., having a direction along which the layer extends the most.

[0017] A conventional vertical dual (or sometimes also called double) MTJ STT-MRAM memory cell comprises a layer stack including a bottom hard magnetization layer, a bottom antiferromagnetic coupling layer (a first antiferromagnetic coupling layer), a bottom reference layer, a first tunnel barrier, a free layer, a second tunnel barrier, a top reference layer, a second antiferromagnetic coupling layer (a second antiferromagnetic coupling layer), and a top hard magnetization layer. The layer stacking direction in the vertical cell is perpendicular to the top surface of the supporting substrate. The magnetization direction of the top reference layer is antiparallel to the magnetization direction of the bottom reference layer. The first tunnel barrier is instrumental in providing a tunneling magnetoresistance (TMR). In a vertical memory cell in which the bottom reference layer has an upward magnetization direction and the top reference layer has a downward magnetization direction, the magnetic state of the free layer can be programmed from an antiparallel to parallel state by inducing flow of electrons from bottom to top, thereby aligning the free layer's magnetization direction to an upward direction. This state is referred to as a parallel state because the magnetization direction of the free layer is parallel to the magnetization direction of the bottom reference layer. The parallel state generates direct spin-transfer torque (STT) on the top reference layer, and generates indirect spin-transfer torque on the bottom reference layer. Such spin-transfer torques have a tendency of flipping the magnetization direction of the top reference layer, thereby rendering the conventional dual reference layer magnetoresistive memory cell inoperable.

[0018] The top reference layer encounters challenges achieving sufficient antiferromagnetic coupling and perpendicular magnetic anisotropy (PMA) for maintaining the downward magnetization direction. In other words, the magnetization direction of the top reference layer in the conventional memory cell is unstable. The spin-transfer torque that the free layer applies to the top magnetization layer is applied in a manner that helps flipping of the magnetization direction of the top reference layer upward, and is aided by stray magnetic fields emanating from the top hard magnetization layer, the free layer, and the bottom reference layer. Thus, maintaining the magnetization direction of the top reference layer in the downward direction is a challenge for the conventional dual MTJ STT-MRAM memory cells.

[0019] Mathematically, the effective field applied to the top reference layer having a downward magnetization from other layers of the layer stack in the conventional dual MTJ STT-MRAM memory cell while the free layer has an upward magnetization direction is given by:

$$h_{\text{sub.eff}}(\text{RL}) = -h_{\text{sub.m,THL}} - h_{\text{sub.m,FL}} - h_{\text{sub.STT,FL}} - h_{\text{sub.m,BRL}} + h_{\text{sub.m,BHL}}$$
, in which $h_{\text{sub.m,THL}}$ represents the effective field due to the magnetization of the top hard magnetization layer, $h_{\text{sub.m,FL}}$ represents the effective field due to the magnetization of the free layer, $h_{\text{sub.STT,FL}}$ represents the effective field due to the spin-torque transfer effect from the free layer, $h_{\text{sub.m,BRL}}$ represents the effective field due to the magnetization of the bottom reference layer, and $h_{\text{sub.m,BHL}}$ represents the effective field due to the magnetization of the bottom hard magnetization layer. The negative signs correspond to a direction that promotes flipping of the downward magnetization of the top reference layer upward, and the positive sign corresponds a direction that suppresses flipping of the downward magnetization of the top reference layer.

Likewise, the effective field applied to the top reference layer having a downward magnetization from other layers of the layer stack in the conventional dual MTJ STT-MRAM memory cell while the free layer has a downward magnetization direction is given by:

$$h_{\text{sub.eff}}(\text{RL}) = -h_{\text{sub.m,THL}} + h_{\text{sub.m,FL}} - h_{\text{sub.STT,FL}} - h_{\text{sub.m,BRL}} + h_{\text{sub.m,BHL}}$$
. Thus, the effective field that induces back-hopping on the top reference layer is the greatest when the

magnetization of the free layer is parallel to the magnetization of the bottom reference layer. The back-hopping of the top reference layer induced by the effective field from other layers in the layer stack can cause a high write error rate during operation of the conventional dual MTJ STT-MRAM memory cell.

[0020] Embodiments of the present disclosure provide a solution for reducing back-hopping by providing a dual MTJ STT-MRAM memory cell in which a top reference layer and a bottom reference layer that are located on opposite sides of a free layer have parallel magnetization direction during the writing of the memory cell. Various aspects of the present disclosure are now described with reference to accompanying figures.

[0021] In one embodiment, one of the two reference layers comprises a negative spin polarization material which provides a stable dual MTJ STT-MRAM memory cell operation in the configuration in which the top reference layer and the bottom reference layer have the same magnetization direction. In one embodiment, the other one of the two reference layers comprises a positive spin polarization material. The parallel alignment of magnetization directions of the two reference layers of the embodiments of the present disclosure mitigates the back-hopping problem in the dual MTJ STT-MRAM memory cell.

[0022] Referring to FIG. 1, a schematic diagram is shown for a magnetoresistive random access memory (MRAM) device **500** including memory cells **180** of any embodiment of the present disclosure in an array configuration. The MRAM device **500** includes an array of memory cells **180**, which may be configured as a two-dimensional array or as a three-dimensional array. As used herein, a “random access memory” (RAM) refers to a memory device containing memory cells that allow random access, e.g., access to any selected memory cell upon a command for reading the contents of the selected memory cell. The MRAM device **500** of the embodiment of the present disclosure is a random access memory device including a magnetoresistive memory element within each memory cell.

[0023] The MRAM device **500** of an embodiment of the present disclosure includes a memory array region **550** containing an array of memory cells **180** located at the intersection of the respective word lines (which may comprise first electrically conductive lines **30** as illustrated or as second electrically conductive lines **90** in an alternate configuration) and bit lines (which may comprise second electrically conductive lines **90** as illustrated or as first electrically conductive lines **30** in an alternate configuration). Each of the memory cells **180** can be a two terminal memory cell including a respective first electrode and a respective second electrode. In one embodiment, the first electrodes can be connected to the first electrically conductive lines **30**, and the second electrodes can be connected to the second electrically conductive lines **90**. Alternatively, the first electrodes can be connected to the second electrically conductive lines **90**, and the first electrodes can be connected to the first electrically conductive lines **30**.

[0024] The MRAM device **500** may also contain a row decoder **560** connected to the word lines, a sense circuitry **570** (e.g., a sense amplifier and other bit line control circuitry) connected to the bit lines, a column decoder **580** connected to the bit lines, and a data buffer **590** connected to the sense circuitry. Multiple instances of the memory cells **180** are provided in an array configuration that forms the MRAM device **500**. It should be noted that the location and interconnection of elements are schematic and the elements may be arranged in a different configuration. Further, a memory cell **180** may be manufactured as a discrete device, i.e., a single isolated device.

[0025] Each memory cell **180** includes two magnetic tunnel junctions including a common free layer. The two magnetic tunnel junctions are provided between a first electrode and a second electrode within each memory cell **180**. In one embodiment, each memory cell **180** can be a spin-transfer torque (STT) magnetoresistive memory cell in which the magnetization of the free layer can be programmed deterministically by bidirectional spin-polarized current that tunnels through two magnetic tunnel junctions. The details of the embodiment memory cell **180** are now described in detail.

[0026] FIG. 2A illustrates a magnetoresistive memory cell **180** of a dual MTJ STT-MRAM device in a parallel free layer state according to a first embodiment of the present disclosure. FIG. 2B illustrates the magnetoresistive memory cell **180** in an antiparallel free layer state according to the first embodiment of the present disclosure. Thus, FIGS. 2A and 2B illustrate the same magnetoresistive memory cell **180** in two different programmed states, i.e., in a parallel free layer state in which the magnetization direction of the free layer is parallel to the magnetization direction of the reference layers, as shown in FIG. 2A, and in an antiparallel free layer state, in which the magnetization direction of the free layer is antiparallel to the magnetization direction of the reference layers, as shown in FIG. 2B.

[0027] The magnetoresistive memory cell **180** can be formed on any support or substrate, such as an insulating support **20** (which may include a silicon oxide layer), and can include a first electrode **32** that may be electrically connected to or comprises a portion of a first electrically conductive line **30** (such as a word line or a bit line) and a second electrode **92** that may be electrically connected to or comprises a portion of a second electrically conductive line **90** (such as a bit line or a word line). The magnetoresistive memory cell **180** further comprises a layer stack (**112, 114, 132, 134, 136, 144, 142, 154, 152**) located between the first electrode **32** and the second electrode **92**. The layer stack (**112, 114, 132, 134, 136, 144, 142, 154, 152**) comprises a dual magnetic tunnel junction (MTJ) **140**, which comprise, from one end to another, a first reference layer **132**, a first tunnel barrier layer **134**, a free layer **136**, a second tunnel barrier layer **144**, a second reference layer **142**.

[0028] In one embodiment, a first one of the first reference layer **132** and the second reference layer **142** comprises a positive spin polarization material, and a second one of the first reference layer **132** and the second reference layer **142** comprises a negative spin polarization material. In one embodiment, the first reference layer **132** comprises a positive spin polarization material, and the second reference layer **142** comprises a negative spin polarization material. Alternatively, the second reference layer **142** comprises a positive spin polarization material, and the first reference layer **132** comprises a negative spin polarization material.

[0029] Positive spin polarization refers to a phenomenon in which the spin direction of electrons passing through a magnetic material aligns with the direction of magnetization of the magnetic material. In other words, the magnetization direction is parallel to the direction of the spin polarized current. A magnetic material that provides positive spin polarization is herein referred to as a positive spin polarization material. Positive spin polarization occurs in most ferromagnetic materials. In contrast, negative spin polarization refers to a phenomenon in which the spin direction of electrons passing through a magnetic material aligns along the opposite direction of the direction of magnetization of the magnetic material. In other words, the magnetization direction is antiparallel (i.e., opposite) to the direction of the spin polarized current. A magnetic material that provides negative spin polarization is herein referred to as a negative spin polarization material. Examples of positive spin polarization materials include iron, cobalt, nickel, and their alloys (e.g., CoFe, CoFeB, NiFe, etc). Examples of negative spin polarization materials include disordered body centered cubic FeCr alloys (e.g., Fe.sub.1-xCr.sub.x with the x value of below 0.7), Mn.sub.2VGa Heusler alloys, Fe.sub.4N and SrRuO.sub.3.

[0030] FIGS. 3A-3D illustrate effects of spin polarization materials on the spin distribution of electrons that pass through a respective spin polarization material. FIG. 3A illustrates the effect of a negative spin polarization material on electrons with random spin that flow downward while the magnetization direction of the negative spin polarization material points upward. FIG. 3B illustrates the effect of a positive spin polarization material on electrons with random spin that flow downward while the magnetization direction of the positive spin polarization material points upward. FIG. 3C illustrates the effect of a negative spin polarization material on electrons with random spin that flow upward while the magnetization direction of the negative spin polarization material points upward. FIG. 3D illustrates the effect of a positive spin polarization material on electrons with random spin that flow upward while the magnetization direction of the positive spin

polarization material points upward.

[0031] Thus, a series connection of a negative spin polarization material and a positive spin polarization material with a free layer therebetween provides spin-polarized electrons in the free layer in either spin direction (i.e., an upward direction or a downward direction) by selecting the direction current flow (i.e., the direction of flow of the electrons), and therefore STT from the two reference layers on the free layer are additive, and the switching current to flip the magnetization of the free layer is reduced. Thus, the use of a negative spin polarization material for one of the first reference layer **132** and the second reference layer **142** in conjunction with use of a positive spin polarization material for the other of the first reference layer **132** and the second reference layer **142** selects the direction of the spin of the electrons (i.e., the direction of the spin polarized current) that passes the free layer **136** between the upward direction and the downward direction in the magnetoresistive memory cell **180** illustrated in FIGS. 2A and 2B.

[0032] Referring back to FIGS. 2A and 2B, the magnetization direction of the second reference layer **142** is parallel to the magnetization direction of the first reference layer **132**. In the magnetoresistive memory cell **180** illustrated in FIGS. 2A and 2B, the magnetization directions of the first reference layer **132** and the second reference layer **142** are downward.

[0033] In one embodiment, the magnetoresistive memory cell **180** comprises a first magnetic polarizer layer **112** located between the first electrode **32** and the first reference layer **132** and configured to stabilize the magnetization direction of the first reference layer **132**. The first magnetic polarizer layer **112** can be any material layer or a material layer stack that can function as a first hard magnetization layer, i.e., a magnetic material layer having a stable magnetization direction. In one embodiment, the first magnetic polarizer layer **112** has a magnetization direction that is antiparallel to the magnetization direction of the first reference layer **132**, and a first antiferromagnetic coupling layer **114** is located between the first magnetic polarizer layer **112** and the first reference layer **132** and provides antiferromagnetic coupling therebetween. In one embodiment, the first magnetic polarizer layer **112** comprises a ferromagnetic multilayer structure including a superlattice, an exchange-bias-inducing antiferromagnetic layer, or a stack of at least one ferromagnetic material layer and at least one antiferromagnetic layer. Alternatively, the first magnetic polarizer layer **112** comprises a hard magnetization layer (i.e., a permanent magnet).

[0034] In a non-limiting illustrative example, the first magnetic polarizer layer **112** may comprise a superlattice of cobalt layers and platinum layers. The number repetitions of a combination of a cobalt layer and a platinum layer may be in a range from 2 to 10, such as from **3** to **6**, although lesser and greater number of repetitions may also be employed. In an illustrative example, the cobalt layers may have a respective thickness of 0.2 nm to 0.5 nm, and the platinum layers may have a respective thickness of about 0.1 nm to 0.3 nm. It is understood that a material layer having a thickness that is less than the thickness of a monolayer refers to a discontinuous layer having a fractional coverage that is equal to the ratio of the thickness of the material layer to the thickness of the monolayer.

[0035] The first antiferromagnetic coupling layer **114** has a material composition and a thickness that provide antiferromagnetic coupling between the first magnetic polarizer layer **112** and the first reference layer **132**. In one embodiment, the first antiferromagnetic coupling layer **114** can include ruthenium or iridium, and can have a thickness in a range from 0.3 nm to 0.8 nm. The combination of the first magnetic polarizer layer **112**, the first antiferromagnetic coupling layer **114** and the first reference layer **132** may comprise a first synthetic antiferromagnetic structure (“SAF”).

[0036] The magnetoresistive memory cell **180** comprises a second magnetic polarizer layer **152** located between the second electrode **92** and the second reference layer **142** and configured to stabilize the magnetization direction of the second reference layer **142**. The second magnetic polarizer layer **152** can be any material layer or a material layer stack that can function as a second hard magnetization layer, i.e., a magnetic material layer having a stable magnetization direction. The second magnetic polarizer layer **152** has a magnetization direction that is antiparallel to the

magnetization direction of the second reference layer **142**, and a second antiferromagnetic coupling layer **154** is located between and contacts each of the second magnetic polarizer layer **152** and the second reference layer **142** and provides antiferromagnetic coupling therebetween. Thus, the magnetization direction of the second magnetic polarizer layer **152** is parallel to the magnetization direction of the first magnetic polarizer layer **112**. The second magnetic polarizer layer **152** may comprise a layer or layer stack having the same composition as the first magnetic polarizer layer **112**. In the magnetoresistive memory cell **180**, the magnetization directions of the first magnetic polarizer layer **112** and the second magnetic polarizer layer **152** are downward.

[0037] In one embodiment, the first tunnel barrier layer **134** is located between (and optionally in contact with) a first surface of the free layer **136** and with a surface of the first reference layer **132**, and the second tunnel barrier layer **144** is located between (and optionally in contact with) a second surface of the free layer **136** and with a surface of the second reference layer **142**. The first and second tunnel barrier layers (**134**, **144**) may comprise MgO or another suitable electrically insulating material.

[0038] In one embodiment, the second reference layer **142** overlies the first reference layer **132**; the first reference layer **132** comprises the positive spin polarization material; and the second reference layer **142** comprises the negative spin polarization material. Alternatively, the second reference layer **142** overlies the first reference layer **132**; the first reference layer **132** comprises the negative spin polarization material; and the second reference layer **142** comprises the positive spin polarization material.

[0039] In one embodiment, the layer stack (**112**, **114**, **132**, **134**, **136**, **144**, **142**, **154**, **152**) can be formed by depositing a continuous layer stack and by patterning the continuous layer stack using an anisotropic etch process or an ion beam etching process. In this case, the layer stack (**112**, **114**, **132**, **134**, **136**, **144**, **142**, **154**, **152**) may be a vertical stack in which the first reference layer **132**, the first tunnel barrier layer **134**, the free layer **136**, the second tunnel barrier layer **144**, the second reference layer **142** are arranged along a vertical direction.

[0040] The magnetoresistive memory cell **180** may be provided in an MRAM array of the STT-MRAM device **500** described with reference to FIG. **1**. In this case, the magnetoresistive memory cell **180** is a memory cell within a two-dimensional array of memory cells; the first electrode **32** contacts or is a portion of a word line (which may be a first electrically conductive line **30** or a second electrically conductive line **90**) for the two-dimensional array of memory cells; and the second electrode **92** contacts or is a portion of a bit line (which may be a second electrically conductive line **90** or a first electrically conductive line **30**) for the two-dimensional array of memory cells.

[0041] In the parallel state of FIG. **2A** in which the magnetizations of the first reference layer **132** (i.e., the bottom reference layer) and the second reference layer **142** (i.e., the top reference layer) point upward and the magnetization of the free layer **136** points upward, the effective field applied to the second reference layer **142** from other layers of the layer stack in the magnetoresistive memory cell of the present disclosure is given by:

$$h_{\text{sub}}.\text{eff}(\text{RL}) = -h_{\text{sub}}.\text{m},\text{THL} + h_{\text{sub}}.\text{m},\text{FL} + h_{\text{sub}}.\text{STT},\text{FL} + h_{\text{sub}}.\text{m},\text{BRL} - h_{\text{sub}}.\text{m},\text{BHL}$$
, in which $h_{\text{sub}}.\text{m},\text{THL}$ represents the effective field due to the magnetization of the top hard magnetization layer **152**, $h_{\text{sub}}.\text{m},\text{FL}$ represents the effective field due to the magnetization of the free layer **136**, $h_{\text{sub}}.\text{STT},\text{FL}$ represents the effective field due to the spin-torque transfer effect from the free layer **136**, $h_{\text{sub}}.\text{m},\text{BRL}$ represents the effective field due to the magnetization of the bottom reference layer **132**, and $h_{\text{sub}}.\text{m},\text{BHL}$ represents the effective field due to the magnetization of the bottom hard magnetization layer **112**. The negative signs correspond to a direction that promotes flipping of the upward magnetization of the second reference layer **142** (i.e., the top reference layer) downward, and the positive sign corresponds a direction that suppresses flipping of the magnetization of the second reference layer **142**.

[0042] Similarly, in the antiparallel state of FIG. **2B** in which the magnetizations of the first

reference layer **132** (i.e., the bottom reference layer) and the second reference layer **142** (i.e., the top reference layer) point upward and the magnetization of the free layer **136** points downward, the effective field applied to the second reference layer **142** from other layers of the layer stack in the magnetoresistive memory cell of the present disclosure is given by:

$$h_{\text{sub}}^{\text{eff}}(\text{RL}) = -h_{\text{sub}}^{\text{m,THL}} - h_{\text{sub}}^{\text{m,FL}} + h_{\text{sub}}^{\text{STT,FL}} + h_{\text{sub}}^{\text{m,BRL}} - h_{\text{sub}}^{\text{m,BHL}}$$
. Thus, for both directions of the magnetization of the free layer **136**, the components of the effective field on the second reference layer **142** (i.e., the top reference layer) tend to cancel out among one another, and thus, back-hopping of the magnetization of the second reference layer **142** is reduced or eliminated in the embodiment magnetoresistive memory cell **180**. Further, for both directions of the magnetization of the free layer **136**, the components of the effective field on the first reference layer **132** (i.e., the bottom reference layer) also tend to cancel out among one another, and thus, back-hopping of the magnetization of the first reference layer **132** is also reduced or eliminated in the embodiment magnetoresistive memory cell **180**. Thus, the frequency of a write error during operation of the embodiment magnetoresistive memory cell **180** can be significantly reduced.

[0043] Thus, in the parallel state of FIG. 2A, the spin transfer torque from the free layer **136** on the second reference layer **142** acts to stabilize the magnetization direction of the second reference layer **142**, with the additional support from stray fields from both the free layer **136** and the first reference layer **132** for stabilizing the magnetization direction of the second reference layer **142**. This reduced or eliminates the back-hopping problem for the top reference layer in the parallel state, and additionally reduces the probability of back-hopping in the antiparallel state. Therefore, use of a negative spin polarization material for one of the first reference layer **132** and the second reference layer **142** and use of a positive spin polarization material for another of the first reference layer **132** and the second reference layer **142** in conjunction with a parallel alignment of magnetization between the first reference layer **132** and the second reference layer **142** can reduce or eliminate back-hopping of the magnetization direction for the second reference layer **142** and also for the first reference layer **132**.

[0044] FIG. 4A illustrates an alternative configuration of a magnetoresistive memory cell **180** in a parallel state according to an embodiment of the present disclosure. FIG. 4B illustrates the alternative configuration of the magnetoresistive memory cell **180** in an antiparallel state according to an embodiment of the present disclosure.

[0045] Referring to FIGS. 4A and 4B, the alternative configuration of the magnetoresistive memory cell **180** can be derived from the magnetoresistive memory cell **180** illustrated in FIGS. 2A and 2B by reversing the magnetization directions of the first magnetic polarizer layer **112**, the first reference layer **132**, the second reference layer **142**, and the second magnetic polarizer layer **152**. Specifically, the magnetization directions of the first reference layer **132** and the second reference layer **142** are downward, and the magnetization direction of the first magnetic polarizer layer **112** and the second magnetic polarizer layer **152** are upward. The alternative configuration of the magnetoresistive memory cell **180** of FIGS. 4A and 4B operates in the same manner as the magnetoresistive memory cell **180** of FIGS. 2A and 2B except that the direction of the effective field on the free layer **136** is reversed in the alternative configuration of the magnetoresistive memory cell **180** of FIGS. 4A and 4B. The mechanism for enhancement of the stability of the magnetization direction of the second reference layer **142** and the magnetization direction of the first reference layer **132** is the same.

[0046] FIGS. 5A-5D illustrate effects of spin polarization materials on the spin distribution of electrons that pass through a respective spin polarization material. FIG. 5A illustrates the effect of a negative spin polarization material on electrons that flow downward while the magnetization direction of the negative spin polarization material points downward. FIG. 5B illustrates the effect of a positive spin polarization material on electrons that flow downward while the magnetization direction of the positive spin polarization material points downward. FIG. 5C illustrates the effect of a negative spin polarization material on electrons that flow upward while the magnetization

direction of the negative spin polarization material points downward. FIG. 5D illustrates the effect of a positive spin polarization material on electrons that flow upward while the magnetization direction of the positive spin polarization material points downward.

[0047] Referring collectively to FIGS. 2A-5D and according to another aspect of the present disclosure, a method of operating a spin transfer torque (STT) magnetoresistive random memory (MRAM) device includes providing a magnetoresistive memory cell **180** which comprises a first electrode **32**, a second electrode **92**, and a layer stack (**112, 114, 132, 134, 136, 144, 142, 154, 152**) located between the first electrode **32** and the second electrode **92** comprises applying a read bias across the first electrode **32** and the second electrode **92** while the magnetization directions of the first reference layer **132** and the second reference layer **142** are parallel to each other; and determining a magnetization direction of the free layer **136** relative to the magnetization directions of the first reference layer **132** and the second reference layer **142** based on a magnitude of electrical current that flows through the layer stack (**112, 114, 132, 134, 136, 144, 142, 154, 152**) under the read bias voltage.

[0048] The magnetization directions of the first reference layer **132** and the second reference layer **142** are parallel to each other during a normal mode of operation for the magnetoresistive memory cell **180**. In one embodiment, the magnetoresistive memory cell **180** may be programmed into a parallel state in which the magnetization direction of the free layer **136** is parallel to the magnetization directions of the first reference layer **132** and the second reference layer **142** by flowing a programming electrical current through the free layer **136** while the magnetization directions of the first reference layer **132** and the second reference layer **142** are parallel to each other. In one embodiment, the magnetoresistive memory cell **180** may be programmed into an antiparallel state in which the magnetization direction of the free layer **136** is antiparallel to the magnetization directions of the first reference layer **132** and the second reference layer **142** by flowing a programming electrical current through the free layer **136** while the magnetization directions of the first reference layer **132** and the second reference layer **142** are parallel to each other.

[0049] In one embodiment, a first one of the first reference layer **132** and the second reference layer **142** comprises a positive spin polarization material; and a second one of the first reference layer **132** and the second reference layer **142** comprises a negative spin polarization material.

[0050] Although the foregoing refers to particular preferred embodiments, it will be understood that the disclosure is not so limited. It will occur to those of ordinary skill in the art that various modifications may be made to the disclosed embodiments and that such modifications are intended to be within the scope of the disclosure. Compatibility is presumed among all embodiments that are not alternatives of one another. The word “comprise” or “include” contemplates all embodiments in which the word “consist essentially of” or the word “consists of” replaces the word “comprise” or “include,” unless explicitly stated otherwise. Whenever two or more elements are listed as alternatives in a same paragraph or in different paragraphs, a Markush group including a listing of the two or more elements is also impliedly disclosed. Whenever the auxiliary verb “can” is employed in this disclosure to describe formation of an element or performance of a processing step, an embodiment in which such an element or such a processing step is not performed is also expressly contemplated, provided that the resulting apparatus or device can provide an equivalent result. As such, the auxiliary verb “can” as applied to formation of an element or performance of a processing step should also be interpreted as “may” or as “may, or may not” whenever omission of formation of such an element or such a processing step is capable of providing the same result or equivalent results, the equivalent results including somewhat superior results and somewhat inferior results. Where an embodiment employing a particular structure and/or configuration is illustrated in the present disclosure, it is understood that the present disclosure may be practiced with any other compatible structures and/or configurations that are functionally equivalent provided that such substitutions are not explicitly forbidden or otherwise known to be impossible to

one of ordinary skill in the art. If publications, patent applications, and/or patents are cited herein, each of such documents is incorporated herein by reference in their entirety.

Claims

1. A device comprising a magnetoresistive memory cell which comprises: a first electrode; a second electrode; and a layer stack located between the first electrode and the second electrode and comprising, from one end to another, a first reference layer, a first tunnel barrier layer, a free layer, a second tunnel barrier layer, and a second reference layer, wherein: a first one of the first reference layer and the second reference layer comprises a positive spin polarization material; a second one of the first reference layer and the second reference layer comprises a negative spin polarization material; and a magnetization direction of the second reference layer is parallel to a magnetization direction of the first reference layer.
2. The device of claim 1, wherein the device comprises a spin transfer torque (STT) magnetoresistive random memory (MRAM) array comprising a plurality of dual magnetic tunnel junction STT-MRAM memory cells.
3. The device of claim 1, wherein the magnetoresistive memory cell further comprises a first magnetic polarizer layer located between the first electrode and the first reference layer.
4. The device of claim 3, further comprising a first antiferromagnetic coupling layer is located between the first magnetic polarizer layer and the first reference layer.
5. The device of claim 4, wherein the first magnetic polarizer layer has a magnetization direction that is antiparallel to the magnetization direction of the first reference layer.
6. The device of claim 4, wherein the first magnetic polarizer layer comprises a hard magnetization layer or a ferromagnetic multilayer structure including a superlattice, an exchange-bias-inducing antiferromagnetic layer, or a stack of at least one ferromagnetic material layer and at least one antiferromagnetic layer.
7. The device of claim 4, wherein the magnetoresistive memory cell further comprises a second magnetic polarizer layer located between the second electrode and the second reference layer.
8. The device of claim 7, further comprising a second antiferromagnetic coupling layer is located between the second magnetic polarizer layer and the second reference layer.
9. The device of claim 7, wherein: the second magnetic polarizer layer has a magnetization direction that is antiparallel to the magnetization direction of the second reference layer; and the magnetization direction of the first magnetic polarizer layer is parallel to the magnetization direction of the second magnetic polarizer layer.
10. The device of claim 1, wherein: the first tunnel barrier layer is in contact with a first surface of the free layer and with a surface of the first reference layer; and the second tunnel barrier layer is in contact with a second surface of the free layer and with a surface of the second reference layer.
11. The device of claim 1, wherein: the second reference layer overlies the first reference layer; the first reference layer comprises the positive spin polarization material; and the second reference layer comprises the negative spin polarization material.
12. The device of claim 1, wherein: the second reference layer overlies the first reference layer; the first reference layer comprises the negative spin polarization material; and the second reference layer comprises the positive spin polarization material.
13. The device of claim 1, wherein: the positive spin polarization material comprises Fe, Co, Ni or an alloy thereof; and the negative spin polarization material comprises $\text{Fe}_{1-x}\text{Cr}_x$ where $x < 0.7$, a $\text{Mn}_{0.2}\text{VGa}$ Heusler alloy, Fe_4N or SrRuO_3 .
14. The device of claim 1, wherein: the layer stack is a vertical stack in which the first reference layer, the first tunnel barrier layer, the free layer, the second tunnel barrier layer, the second reference layer are arranged along a vertical direction; and the layer stack is located over a support.
15. A method of operating a spin transfer torque (STT) magnetoresistive random memory (MRAM)

device, comprising: providing a magnetoresistive memory cell which comprises a first electrode, a second electrode, and a layer stack located between the first electrode and the second electrode and comprising, from one end to another, a first reference layer, a first tunnel barrier layer, a free layer, a second tunnel barrier layer, a second reference layer, wherein the first reference layer and the second reference layer have magnetization directions that are parallel to each other; applying a read bias across the first electrode and the second electrode while the magnetization directions of the first reference layer and the second reference layer are parallel to each other; and determining a magnetization direction of the free layer relative to the magnetization directions of the first reference layer and the second reference layer based on a magnitude of electrical current that flows through the layer stack under the read bias.

16. The method of claim 15, further comprising programming the magnetoresistive memory cell into a parallel state in which the magnetization direction of the free layer is parallel to the magnetization directions of the first reference layer and the second reference layer by flowing a programming electrical current through the free layer while the magnetization directions of the first reference layer and the second reference layer are parallel to each other.

17. The method of claim 15, further comprising programming the magnetoresistive memory cell into an antiparallel state in which the magnetization direction of the free layer is antiparallel to the magnetization directions of the first reference layer and the second reference layer by flowing a programming electrical current through the free layer while the magnetization directions of the first reference layer and the second reference layer are parallel to each other.

18. The method of claim 15, wherein: a first one of the first reference layer and the second reference layer comprises a positive spin polarization material; and a second one of the first reference layer and the second reference layer comprises a negative spin polarization material.

19. The method of claim 18, wherein: the positive spin polarization material comprises Fe, Co, Ni or an alloy thereof; and the negative spin polarization material comprises $\text{Fe}_{1-x}\text{Cr}_x$ where $x < 0.7$, a $\text{Mn}_{0.2}\text{VGa}$ Heusler alloy, $\text{Fe}_{0.4}\text{N}$ or SrRuO_3 .

20. The method of claim 18, wherein the magnetoresistive memory cell further comprises: a first magnetic polarizer layer located between the first electrode and the first reference layer; a first antiferromagnetic coupling layer is located between the first magnetic polarizer layer and the first reference layer; a second magnetic polarizer layer located between the second electrode and the second reference layer; and a second antiferromagnetic coupling layer is located between the second magnetic polarizer layer and the second reference layer.
